



BIOLOGY 4

ACADEMIC

Grade 12

Curriculum Guide

BIOLOGY 4

Prerequisites: Biology 1, Biology 2, Biology 3

Course Description:

Biology 4 builds on the concepts and scientific skills developed in Biology 3 by examining the classification of organisms and their interactions within ecosystems, as well as the role of science and technology in addressing environmental and societal challenges. The learners explore the principles of taxonomy, biodiversity conservation, and ecological interactions, including energy flow and the interconnections among terrestrial and aquatic ecosystems. Through these topics, they develop an understanding of how organisms adapt to their environments and how ecosystems maintain balance.

The course also introduces key concepts in environmental science, biotechnology, and artificial intelligence. The learners analyze how human activities influence ecosystems, evaluate sustainable practices such as Circular and Blue Economy approaches, and examine the applications and ethical implications of emerging technologies. The learners analyze environmental issues, evaluate technological solutions, and propose informed actions that support biodiversity conservation and sustainable development.

Elective: Academic

Time Allotment: 80 hours for one term

CONTENT	CONTENT STANDARDS <i>The learners demonstrate understanding that...</i>	LEARNING COMPETENCIES <i>The learners...</i>
Taxonomy and Ecosystem		
1. Taxonomy: <i>Identification, Nomenclature, and Classification</i>	1. organisms are classified based on shared structural characteristics and molecular evidence, and that biological classification reflects	1. describe the contributions of important scientists to the development of biological taxonomy;
		2. explain how structural characteristics and molecular evidence, such as DNA sequences, are used in the classification of living organisms;

CONTENT	CONTENT STANDARDS <i>The learners demonstrate understanding that...</i>	LEARNING COMPETENCIES <i>The learners...</i>
	evolutionary relationships and adaptations of organisms;	
		3. explain how the classification of organisms relates to their adaptations to different environments;
		4. describe the distinctive attributes of a specific taxon relative to other taxa;
		5. classify organisms into a specific taxon based on its distinctive characteristics;
	2. organisms can be identified and classified to support biodiversity conservation and the prioritization of species that are extinct or at risk of extinction;	6. use information from secondary sources to: a. identify extinct species and explain the ecological impacts of their loss in the environment (e.g., dodo, Philippine rhinoceros); and b. identify species at risk of extinction and explain the importance of conservation prioritization (e.g., terrestrial species such as the Philippine eagle and tamaraw; aquatic and marine species such as the Philippine crocodile, hawksbill sea turtle, tabios/tawilis, and giant clam; and threatened tree species such as narra and molave);
2. Interaction and Interdependence	3. organisms interact with one another and with abiotic factors within ecosystems; and	7. explain ecological relationships among organisms in an ecosystem, including competition, predation, mutualism, commensalism, and parasitism;
		8. explain how organisms adapt to abiotic factors such as climate change;
		9. explain the basic principles and components of ecosystems;
	4. energy flows through ecosystems and terrestrial and aquatic ecosystems are interconnected,	10. use diagrams to trace the flow of energy in an ecosystem (e.g., food chain, food web, and ecological pyramid); and

CONTENT	CONTENT STANDARDS <i>The learners demonstrate understanding that...</i>	LEARNING COMPETENCIES <i>The learners...</i>
	contributing to ecosystem balance.	11. explain how terrestrial and aquatic ecosystems are interconnected through interactions among forests, rivers, mangroves, seagrass beds, and coral reefs.
Performance Standards <i>The learners can classify organisms based on their distinctive characteristics, explain ecological interactions and energy flow within ecosystems, and present evidence-based explanations of how biodiversity conservation and ecosystem interconnections contribute to environmental sustainability.</i>		
Suggested Performance Tasks • Create a visual model or infographic illustrating major nutrient cycles (e.g., water, carbon dioxide–oxygen, nitrogen, phosphorus) within an ecosystem and explain how human activities such as pollution, deforestation, and the introduction of invasive species disrupt these cycles which includes sustainable practices based on Circular or Blue Economy principles that help restore or protect the ecosystem. • Conduct investigation of a real-world application of biotechnology or artificial intelligence (e.g., drug discovery and design, vaccine development, AI-assisted disease diagnosis, or biodiversity monitoring).		
Environmental Science, Biotechnology, and Artificial Intelligence		
3. Environmental Science	5. human activities and development interventions can positively and negatively impact ecosystems and natural resources;	12. use diagrams to explain the nutrient cycles in ecosystems, including the water cycle, carbon dioxide–oxygen cycle, phosphorus cycle, and nitrogen cycle;
		13. describe different ways human activities affect ecosystems, such as pollution, deforestation, and the introduction of invasive species;
		14. explain how the principles of Circular and Blue Economy promote proper management of wastes and natural resources and contribute to a healthy ecosystem and human health (e.g., recycling and composting of wastes, sustainable fisheries, mangrove conservation, and seaweed farming);

CONTENT	CONTENT STANDARDS <i>The learners demonstrate understanding that...</i>	LEARNING COMPETENCIES <i>The learners...</i>
4. Biotechnology and Artificial Intelligence	6. biotechnology and artificial intelligence contribute to improvements in health, food security, and environmental sustainability; and	15. evaluate the environmental impacts of development projects using environmental impact assessment (EIA) to support sustainable decision-making;
	7. the development and application of technologies such as biotechnology and artificial intelligence (AI) involve ethical considerations and social, legal, and environmental trade-offs.	16. explain the importance of biotechnology and artificial intelligence in addressing challenges related to health, food security, and environmental sustainability (e.g., vaccine development, genetically modified crops, disease detection, and environmental monitoring);
		17. explain ethical responsibilities of humans toward the natural environment by examining ethical issues related to human interventions such as genetic modification, cloning, biodiversity conservation, and ecosystem management; and
		18. use information from secondary sources to explain legal, environmental, and social issues related to biotechnology and artificial intelligence, such as genetic modification and cloning, AI-assisted drug discovery and disease diagnosis, and the use of AI in monitoring biodiversity and managing biological data.
Performance Standards <i>The learners can analyze how human activities and technological developments affect ecosystems and natural resources by explaining nutrient cycles, evaluating environmental impacts of development projects, and presenting informed perspectives on the roles, benefits, and ethical implications of biotechnology and artificial intelligence in addressing environmental and societal challenges.</i>		
Suggested Performance Tasks Conduct an investigation of alternative practices or medicines for possible biotechnology applications.		

CONTENT	CONTENT STANDARDS <i>The learners demonstrate understanding that...</i>	LEARNING COMPETENCIES <i>The learners...</i>
• Conduct investigation of a local ecosystem (e.g., forest, river, mangrove, coastal area, or urban green space). The learners identify and classify selected organisms, describe ecological relationships among them, and illustrate energy flow using a food chain or food web. • Explain the impact of an extinct or endangered species on the ecosystem using presentation tools.		